

Advanced Medical Testing Lab

Complete Service Catalog

About Our Laboratory

Advanced Medical Testing Lab is a state-of-the-art diagnostic center committed to providing accurate, reliable, and timely results. With internationally accredited equipment and a team of expert pathologists, radiologists, and technologists, we ensure the highest standards in diagnostic testing.

Our Commitment:

- Accurate and reliable diagnostic results
 - Internationally accredited equipment and processes
 - Expert medical professionals and technologists
 - Patient-focused care with comprehensive support services
 - 24/7 availability for emergency diagnostics
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1. PATHOLOGY SERVICES

Hematology Tests

Complete Blood Count (CBC)

- Components: RBC count, WBC count, Platelet count, Hemoglobin, Hematocrit, MCV, MCH, MCHC, Differential count
- Purpose: Comprehensive blood analysis to detect anemia, infections, blood cancers, immune system disorders
- Sample Required: 2ml blood in EDTA tube
- Fasting Required: No
- Report Delivery: 2-4 hours
- Clinical Applications: Routine health screening, pre-surgical evaluation, monitoring treatment response

Erythrocyte Sedimentation Rate (ESR)

- Purpose: Measures inflammation levels in the body
- Clinical Applications: Monitor autoimmune diseases, infections, malignancies, treatment response

- Sample Required: 2ml blood in citrate tube
- Report Delivery: 2 hours
- Special Notes: Age and gender-specific reference ranges provided

Coagulation Profile

- Tests Included: Prothrombin Time (PT), Partial Thromboplastin Time (PTT), International Normalized Ratio (INR)
- Purpose: Comprehensive assessment of blood clotting function
- Clinical Applications: Pre-surgical screening, anticoagulant therapy monitoring, bleeding disorder evaluation
- Sample Required: 2ml blood in citrate tube
- Report Delivery: 3 hours
- Special Instructions: Avoid medications affecting coagulation 24 hours prior to test

Blood Grouping & Rh Typing

- Tests: ABO grouping, Rh factor determination
- Additional Services: Cross-matching for transfusion preparation
- Purpose: Blood type identification for transfusion, pregnancy planning, organ transplantation
- Sample Required: 2ml blood in EDTA tube
- Report Delivery: 1 hour
- Special Features: Emergency blood grouping available 24/7

Biochemistry Tests

Liver Function Tests (LFT)

- Parameters: ALT, AST, Total Bilirubin, Direct Bilirubin, Alkaline Phosphatase, Albumin, Total Protein, A/G ratio
- Purpose: Comprehensive evaluation of liver health and function
- Clinical Applications: Detect hepatitis, cirrhosis, liver damage, drug toxicity, metabolic liver disease
- Sample Required: 3ml serum
- Fasting Required: 8-12 hours
- Report Delivery: 4-6 hours
- Additional Services: Hepatitis screening correlation available

Kidney Function Tests (KFT)

- Parameters: Serum Creatinine, Blood Urea Nitrogen (BUN), Uric Acid, Sodium, Potassium, Chloride, estimated GFR
- Purpose: Assess kidney filtration capacity and electrolyte balance
- Clinical Applications: Monitor chronic kidney disease, hypertension effects, drug nephrotoxicity
- Sample Required: 3ml serum
- Fasting Required: 8 hours recommended
- Report Delivery: 4-6 hours
- Special Features: Automatic GFR calculation with age and gender adjustment

Lipid Profile

- Parameters: Total Cholesterol, HDL Cholesterol, LDL Cholesterol, Triglycerides, VLDL, TC/HDL ratio, LDL/HDL ratio
- Purpose: Comprehensive cardiovascular risk assessment
- Clinical Applications: Heart disease prevention, diabetes monitoring, treatment effectiveness evaluation
- Sample Required: 3ml serum
- Fasting Required: 12-14 hours mandatory
- Report Delivery: 6 hours
- Additional Services: Cardiovascular risk stratification report included

Glucose Tolerance Test (GTT)

- Method: 2-hour oral glucose tolerance test with standardized glucose load
- Parameters: Fasting glucose, 1-hour post-glucose, 2-hour post-glucose
- Purpose: Definitive diagnosis of diabetes and pre-diabetes conditions
- Sample Required: 2ml blood at 0, 1, and 2-hour intervals
- Preparation Required: 3-day normal carbohydrate diet, 8-12 hour fasting
- Test Duration: 2 hours in-clinic monitoring
- Report Delivery: Same day
- Special Services: Pregnancy GTT protocols available

HbA1c (Glycated Hemoglobin)

- Purpose: Long-term blood sugar control assessment (2-3 month average)
- Clinical Applications: Diabetes diagnosis, treatment monitoring, complications prevention
- Sample Required: 2ml blood in EDTA tube
- Fasting Required: None

- Report Delivery: 4 hours
- Quality Features: HPLC method with international standardization

Microbiology Tests

Culture & Sensitivity Testing

- Types Available: Blood, urine, stool, sputum, wound, throat, cerebrospinal fluid cultures
- Purpose: Identify pathogenic organisms and determine appropriate antibiotic treatment
- Advanced Features: Automated identification systems, rapid organism detection
- Sample Requirements: Varies by specimen type, sterile collection protocols provided
- Report Delivery: 24-72 hours (preliminary results), 5-7 days (complete sensitivity)
- Special Services: ESBL detection, MRSA screening, fungal cultures

Urine Culture & Sensitivity

- Purpose: Comprehensive urinary tract infection diagnosis
- Includes: Colony count quantification, organism identification, complete antibiotic sensitivity panel
- Sample Required: Mid-stream clean catch urine (20ml minimum)
- Collection Instructions: Detailed sterile collection protocol provided
- Report Delivery: 24-48 hours
- Quality Features: Automated processing with manual verification

Stool Culture & Parasitology

- Purpose: Detect bacterial pathogens and parasites causing gastrointestinal illness
- Organisms Detected: Salmonella, Shigella, Campylobacter, E. coli, parasites, ova and cysts
- Sample Required: Fresh stool specimen (2-5g) in appropriate transport medium
- Special Instructions: Multiple samples may be required for parasite detection
- Report Delivery: 3-5 days for bacterial culture, 2-3 days for parasites
- Additional Services: Clostridium difficile toxin testing available

Blood Culture

- Purpose: Detect life-threatening bloodstream infections (septicemia)
- Method: Automated blood culture system with continuous monitoring
- Sample Required: 10ml blood in specialized blood culture bottles
- Collection Protocol: Strict sterile technique with skin antisepsis
- Report Delivery: 24-72 hours with immediate notification for positive results

- Emergency Protocol: Stat processing available for critical patients

Immunology & Serology

Hepatitis Screening Panel

- Tests Included: HBsAg, Anti-HBc (total and IgM), Anti-HBs, Anti-HCV (total), HBeAg, Anti-HBe
- Purpose: Comprehensive hepatitis B and C infection screening and immunity assessment
- Clinical Applications: Blood donor screening, pre-vaccination assessment, chronic infection monitoring
- Sample Required: 3ml serum
- Report Delivery: 4-6 hours
- Advanced Testing: Quantitative viral load testing available upon request

HIV Testing

- Method: 4th generation ELISA detecting both antibodies and p24 antigen
- Confirmatory Testing: Western blot available for positive results
- Sample Required: 2ml serum
- Confidentiality: Strict privacy protocols and counseling services available
- Report Delivery: 4-6 hours
- Window Period: Reduced to 2-4 weeks with 4th generation testing
- Support Services: Pre and post-test counseling available

Dengue Diagnostic Panel

- Tests: NS1 antigen, IgM and IgG antibodies, platelet count
- Purpose: Early dengue detection and disease phase determination
- Clinical Value: Distinguishes primary from secondary infections
- Sample Required: 2ml serum
- Report Delivery: 1-2 hours for rapid tests
- Additional Services: Dengue viral load testing available

Typhoid Testing

- Tests Available: Widal test, Typhidot, blood culture for *Salmonella typhi*
- Purpose: Typhoid fever diagnosis with improved accuracy
- Limitations: Widal test interpretation requires clinical correlation
- Sample Required: 2ml serum for serology, 10ml blood for culture
- Report Delivery: 2-3 hours for serology, 3-5 days for culture

- Recommendation: Combined testing approach for optimal diagnosis

Histopathology Services

Tissue Biopsy Evaluation

- Specimen Types: Surgical specimens, needle biopsies, endoscopic biopsies, cytological samples
- Purpose: Definitive cancer diagnosis, inflammatory condition assessment, treatment planning
- Processing: Routine H&E staining with special stains as clinically indicated
- Sample Requirements: Formalin-fixed tissue specimens with proper labeling
- Report Delivery: 3-5 working days for routine cases
- Expertise: Board-certified pathologists with subspecialty training
- Rush Services: 24-hour reporting available for urgent cases

Immunohistochemistry (IHC)

- Purpose: Specific protein detection for precise cancer subtyping and treatment selection
- Applications: Hormone receptor testing, HER2 evaluation, tumor origin identification
- Quality Assurance: Controls included with every batch, participation in external quality programs
- Report Delivery: 5-7 working days
- Clinical Integration: Correlation with molecular testing when indicated

Cytopathology Services

- Services: Pap smears, fine needle aspirations, fluid cytology, sputum cytology
 - Screening Programs: Cervical cancer screening with HPV co-testing
 - Quality Features: Liquid-based cytology with computer-assisted screening
 - Report Delivery: 2-3 working days
 - Follow-up: Abnormal result counseling and management recommendations
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2. RADIOLOGY & IMAGING SERVICES

Digital X-Ray Services

Comprehensive X-Ray Imaging

- Technology: Digital radiography with immediate image processing
- Services Available: Chest, abdominal, bone, joint, and specialized views
- Image Quality: High-resolution digital images with enhancement capabilities

- Radiation Safety: Low-dose protocols with lead shielding protection
- Report Delivery: 30 minutes with radiologist interpretation
- Emergency Services: 24/7 availability for trauma and urgent cases
- Patient Comfort: Modern positioning aids and efficient workflow

Chest X-Ray

- Standard Views: PA (posterior-anterior) and lateral projections
- Clinical Applications: Lung diseases, heart size assessment, bone abnormalities, pleural conditions
- Preparation: Remove all metal objects and jewelry from chest area
- Special Protocols: Portable bedside X-rays available for hospitalized patients
- Quality Features: Automatic exposure control and image optimization

Bone & Joint X-Rays

- Comprehensive Coverage: All skeletal regions with specialized positioning
- Applications: Fracture detection, arthritis evaluation, bone infections, growth assessments
- Multiple Views: Standard and specialized projections as clinically indicated
- Pediatric Protocols: Age-appropriate techniques with radiation dose optimization
- Weight-bearing Studies: Available for joint evaluation under load conditions

Ultrasound & Doppler Services

Abdominal Ultrasound

- Organs Evaluated: Liver, gallbladder, kidneys, spleen, pancreas, abdominal vessels
- Preparation Required: 6-8 hour fasting for gallbladder and pancreas evaluation
- Clinical Applications: Organ size assessment, stone detection, mass evaluation, fluid collection
- Advanced Features: Elastography for liver fibrosis assessment
- Report Delivery: 1 hour with detailed measurements and findings

Obstetric & Gynecological Ultrasound

- Services: Pregnancy monitoring, pelvic organ evaluation, fertility assessments
- Technology: 3D/4D imaging capabilities for enhanced visualization
- Trimester-Specific Protocols: Growth monitoring, anomaly screening, Doppler studies
- Patient Options: Transabdominal and transvaginal approaches as appropriate
- Family Involvement: Real-time viewing with printed images provided
- Specialized Studies: Nuchal translucency, biophysical profiles, cervical length measurements

Vascular Doppler Studies

- Applications: Carotid arteries, renal vessels, peripheral vascular assessment
- Technology: Color Doppler and spectral analysis capabilities
- Clinical Value: Blood flow quantification, stenosis detection, thrombosis evaluation
- Non-invasive Alternative: Comprehensive vascular assessment without contrast injection
- Specialized Protocols: Deep vein thrombosis screening, arterial insufficiency evaluation

CT Scan Services

Multi-Slice CT Technology

- Equipment: Latest generation CT scanners with rapid acquisition
- Reconstruction: Advanced image processing with multiplanar reconstructions
- Contrast Protocols: IV and oral contrast administration as clinically indicated
- Radiation Optimization: Dose reduction protocols while maintaining diagnostic quality
- Emergency Access: 24/7 availability for trauma and acute conditions

CT Brain Imaging

- Indications: Stroke evaluation, trauma assessment, tumor detection, seizure workup
- Contrast Options: Non-contrast and contrast-enhanced studies available
- Advanced Techniques: CT angiography and perfusion studies
- Preparation: Minimal preparation required, contrast protocols when indicated
- Rapid Results: Critical findings immediately communicated to referring physicians

CT Chest

- Applications: Pulmonary embolism detection, lung cancer screening, interstitial lung disease
- High-Resolution Options: Specialized protocols for lung parenchyma evaluation
- Cardiac Gating: Available for cardiac and aortic evaluation
- Contrast Enhancement: Pulmonary angiography protocols for embolism detection

CT Abdomen & Pelvis

- Comprehensive Evaluation: Complete abdominal and pelvic organ assessment
- Contrast Protocols: Oral and IV contrast for optimal organ visualization
- Emergency Protocols: Trauma evaluation with rapid acquisition techniques
- Specialized Studies: CT colonography, CT urography, hepatic protocols

MRI Services

Advanced MRI Technology

- Equipment: High-field MRI systems with advanced coil technology
- Patient Comfort: Wide bore design with ambient lighting and music
- Safety Protocols: Comprehensive screening for metallic implants and devices
- Contrast Agents: Gadolinium-based agents with safety monitoring
- Specialized Sequences: Functional, diffusion, and perfusion imaging capabilities

MRI Brain & Neurological Imaging

- Applications: Detailed brain structure evaluation, multiple sclerosis, stroke assessment
- Advanced Sequences: Diffusion-weighted imaging, FLAIR, susceptibility-weighted imaging
- Functional Studies: fMRI for pre-surgical planning
- Contrast Enhancement: Gadolinium administration for lesion characterization
- Pediatric Protocols: Sedation services available for young children

MRI Spine

- Coverage: Cervical, thoracic, and lumbar spine regions
- Applications: Disc disease, spinal stenosis, cord pathology, tumor evaluation
- Positioning: Comfortable positioning with surface coils for optimal image quality
- Specialized Studies: Myelography, dynamic studies for instability assessment

MRI Musculoskeletal

- Joint Evaluation: Knee, shoulder, ankle, wrist, and other joint assessments
- Soft Tissue Detail: Superior visualization of ligaments, cartilage, muscles, and tendons
- Sports Medicine: Specialized protocols for athletic injuries
- Arthrography: Direct joint injection techniques for enhanced visualization

Specialized Imaging

Digital Mammography

- Technology: Full-field digital mammography with tomosynthesis (3D mammography)
- Screening Protocols: Evidence-based guidelines for different age groups
- Diagnostic Mammography: Problem-solving for palpable abnormalities
- Quality Assurance: ACR accreditation with rigorous quality control
- Patient Comfort: Compression optimization and positioning expertise

- **Rapid Results:** Same-day reporting for diagnostic studies

DEXA Bone Densitometry

- **Purpose:** Osteoporosis screening and monitoring treatment response
 - **Sites Measured:** Lumbar spine and hip with forearm backup sites
 - **Technology:** Dual-energy X-ray absorptiometry with precision monitoring
 - **Clinical Applications:** Fracture risk assessment, treatment monitoring
 - **Patient Preparation:** Minimal preparation with comfortable positioning
 - **Results:** T-scores and Z-scores with WHO classification
 - **Follow-up Recommendations:** Evidence-based monitoring intervals
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3. CARDIAC DIAGNOSTICS

Non-Invasive Cardiac Testing

Electrocardiogram (ECG)

- **Technology:** 12-lead digital ECG with computer analysis
- **Applications:** Heart rhythm assessment, ischemia detection, electrolyte abnormalities
- **Immediate Results:** Real-time interpretation with cardiologist review
- **Exercise Tolerance:** Stress ECG protocols available
- **Ambulatory Monitoring:** 24-48 hour Holter monitoring services
- **Emergency Protocol:** Stat ECG interpretation for acute presentations

Echocardiography (2D Echo)

- **Comprehensive Assessment:** Complete cardiac structure and function evaluation
- **Doppler Studies:** Blood flow analysis across all cardiac valves
- **Advanced Measurements:** Ejection fraction, wall motion analysis, valve gradients
- **3D Capabilities:** Three-dimensional reconstruction for complex cases
- **Stress Echocardiography:** Exercise and pharmacological stress testing
- **Transesophageal Option:** TEE available for detailed posterior structure evaluation

Treadmill Stress Testing (TMT)

- **Purpose:** Exercise-induced cardiac ischemia detection
- **Protocol:** Bruce protocol with continuous ECG and blood pressure monitoring
- **Safety Measures:** Emergency equipment and trained personnel available

- Contraindication Screening: Comprehensive pre-test evaluation
- Symptom Correlation: Exercise capacity and symptom reproduction assessment
- Recovery Monitoring: Extended post-exercise observation period

Holter Monitoring

- Duration Options: 24-hour, 48-hour, and extended monitoring available
- Technology: Digital recording with arrhythmia detection algorithms
- Patient Diary: Symptom correlation with rhythm abnormalities
- Activity Monitoring: Exercise and daily activity correlation
- Comprehensive Analysis: Heart rate variability and rhythm pattern analysis
- Patient Education: Proper device use and diary maintenance instruction

Advanced Cardiac Imaging

CT Coronary Angiography

- Technology: 256-slice CT with cardiac gating
- Clinical Applications: Non-invasive coronary artery evaluation
- Patient Selection: Appropriate risk stratification for optimal results
- Contrast Protocol: IV contrast injection with heart rate optimization
- Radiation Dose: Optimized protocols for minimum radiation exposure
- Calcium Scoring: Combined coronary calcium assessment
- 3D Reconstruction: Advanced post-processing for comprehensive evaluation

4. MOLECULAR & GENETIC TESTING

PCR & Molecular Diagnostics

COVID-19 RT-PCR Testing

- Method: Real-time reverse transcription PCR
- Sample Collection: Nasopharyngeal and oropharyngeal swabs
- Turnaround Time: 6-12 hours with same-day results available
- Travel Certificates: Official documentation for international travel
- Quality Assurance: WHO-approved protocols with internal controls
- Safety Protocols: Biosafety level 2+ laboratory processing

Tuberculosis Molecular Testing

- Technology: GeneXpert system for rapid TB detection
- Resistance Testing: Rifampicin resistance detection included
- Sample Types: Sputum, pleural fluid, cerebrospinal fluid, lymph node aspirates
- Clinical Advantage: Results available in 4-6 hours versus weeks for culture
- Quality Features: Automated processing with minimal technologist handling
- Treatment Guidance: Immediate resistance information for appropriate therapy selection

Viral Load Quantification

- Applications: HIV, Hepatitis B, Hepatitis C quantitative assessment
- Clinical Use: Treatment monitoring and therapy effectiveness evaluation
- Technology: Real-time PCR with international standardization
- Sample Requirements: Plasma separation within 6 hours of collection
- Reporting: Log values with treatment threshold interpretations
- Monitoring Protocols: Serial testing for treatment response assessment

Cancer Biomarkers

Prostate Specific Antigen (PSA)

- Tests Available: Total PSA, free PSA, PSA ratio calculation
- Clinical Applications: Prostate cancer screening and monitoring
- Age-Specific References: Adjusted normal ranges for different age groups
- Sample Requirements: Serum collection with specific pre-analytical requirements
- Interfering Factors: Guidance on medications and procedures affecting results
- Follow-up Protocols: Abnormal result management recommendations

Tumor Marker Panel

- Markers Available: CEA, CA-125, CA 19-9, AFP, PSA, CA 15-3, HCG
- Clinical Applications: Cancer screening, treatment monitoring, recurrence detection
- Interpretation Services: Clinical correlation with imaging and pathology
- Serial Monitoring: Trend analysis for treatment response assessment
- Quality Assurance: Participation in external quality assessment programs

Genetic Testing Services

Hereditary Cancer Screening

- Gene Panels: BRCA1/BRCA2, Lynch syndrome, familial adenomatous polyposis

- Technology: Next-generation sequencing with variant interpretation
- Genetic Counseling: Pre and post-test counseling services available
- Family Testing: Cascade screening for identified mutations
- Result Interpretation: Clinical significance assessment with actionable recommendations
- Confidentiality: Strict genetic privacy protocols

Prenatal Genetic Testing

- Non-Invasive Prenatal Testing (NIPT): Cell-free fetal DNA analysis
- Screening Coverage: Trisomy 21, 18, 13, and sex chromosome abnormalities
- Gestational Age: Testing available from 9 weeks gestation
- Sample Collection: Simple maternal blood draw
- Result Delivery: 7-10 business days with genetic counseling available
- Follow-up: Confirmatory testing recommendations for positive results

Newborn Screening

- Comprehensive Panel: Metabolic, endocrine, and genetic disorders
 - Sample Collection: Heel stick blood spot on filter paper
 - Timing: Optimal collection 24-72 hours after birth
 - Rapid Processing: Results available within 5-7 days
 - Critical Results: Immediate notification for time-sensitive conditions
 - Follow-up Care: Coordination with pediatric specialists for abnormal results
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5. PREVENTIVE HEALTH PACKAGES

Comprehensive Health Screening Programs

Basic Health Package

- Target Population: Adults 18-40 years, routine health maintenance
- Laboratory Tests: CBC with ESR, fasting glucose, lipid profile, liver function tests (5 parameters), kidney function tests (4 parameters), urine routine examination
- Imaging: Chest X-ray
- Clinical Consultation: Health assessment with preventive care recommendations
- Report Format: Comprehensive health summary with risk factor identification
- Follow-up: Personalized health maintenance recommendations

Executive Health Package

- Target Population: Professionals 35-55 years, comprehensive screening
- Enhanced Testing: All basic package tests plus HbA1c, complete thyroid profile (T3, T4, TSH), vitamin D3, vitamin B12 levels
- Cardiac Evaluation: Resting ECG with interpretation
- Imaging: Abdominal ultrasound for organ assessment
- Medical Consultation: Detailed physician evaluation with lifestyle counseling
- Value-Added Services: Digital health record maintenance

Premium Full Body Check-up

- Target Population: Adults 45+ years, executive-level comprehensive screening
- Advanced Diagnostics: All executive package components plus specialized cardiac and cancer screening
- Cardiac Assessment: 2D echocardiography, treadmill stress test
- Advanced Imaging: CT scan (chest or abdomen based on risk factors)
- Cancer Screening: Age and gender-appropriate tumor markers (PSA/CA-125)
- Hormonal Evaluation: Comprehensive hormone panel assessment
- Bone Health: DEXA scan for osteoporosis screening
- Multi-Specialist Consultations: Internal medicine, cardiology, and appropriate specialty consultations

Women's Wellness Package

- Age-Stratified Programs: Customized packages for 20-35, 36-50, and 50+ age groups
- Gynecological Screening: Pap smear with HPV co-testing, pelvic examination
- Breast Health: Mammography (age-appropriate), breast ultrasound
- Hormonal Assessment: FSH, LH, estrogen, progesterone, testosterone levels
- Reproductive Health: Fertility assessment options, menopause management
- Bone Health: DEXA scan and bone metabolism markers
- Cancer Screening: Cervical, breast, and ovarian cancer markers
- Specialized Consultations: Gynecology and women's health specialist evaluation

Senior Citizen Package

- Target Population: Adults 60+ years, age-specific health concerns
- Comprehensive Metabolic Assessment: Complete chemistry panel with diabetes screening
- Cardiac Evaluation: ECG, echocardiography, stress testing (if appropriate)
- Bone Health: DEXA scan with fracture risk assessment

- Gender-Specific Screening: Prostate evaluation (men), gynecological assessment (women)
 - Cognitive Assessment: Basic cognitive screening tools
 - Pulmonary Function: Spirometry for respiratory health assessment
 - Sensory Evaluation: Vision and hearing screening coordination
 - Geriatric Consultation: Age-appropriate health maintenance and medication review
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6. SPECIALIZED DISEASE PROFILES

Metabolic Disease Management

Comprehensive Diabetes Profile

- Diagnostic Tests: Fasting glucose, HbA1c, postprandial glucose levels
- Metabolic Assessment: Insulin levels, C-peptide, glucose tolerance testing
- Complication Screening: Microalbumin, diabetic nephropathy markers
- Cardiovascular Risk: Complete lipid profile with ratios
- Monitoring Tools: Continuous glucose monitoring coordination
- Patient Education: Diabetes management counseling and lifestyle modification

Thyroid Function Comprehensive Profile

- Complete Assessment: TSH, T3, T4, Free T3, Free T4 levels
- Autoimmune Evaluation: Anti-TPO antibodies, thyroglobulin antibodies
- Clinical Correlation: Symptoms assessment with laboratory findings
- Treatment Monitoring: Therapeutic drug monitoring for thyroid medications
- Specialized Testing: Reverse T3, thyroglobulin for specific indications
- Endocrine Consultation: Specialist referral coordination when indicated

Organ-Specific Health Profiles

Comprehensive Kidney Profile

- Function Assessment: Complete kidney function tests with estimated GFR
- Early Detection: Microalbumin, protein/creatinine ratio
- Advanced Markers: Cystatin C for precise GFR estimation
- Electrolyte Balance: Complete electrolyte panel with phosphorus
- Structural Assessment: Renal ultrasound for anatomical evaluation
- Risk Stratification: Chronic kidney disease staging and progression monitoring

Liver Health Comprehensive Profile

- Function Evaluation: Complete liver function tests with protein synthesis markers
- Viral Screening: Hepatitis A, B, C comprehensive panel
- Metabolic Assessment: Iron studies, ceruloplasmin, alpha-1 antitrypsin
- Tumor Screening: Alpha fetoprotein and other liver cancer markers
- Imaging Correlation: Abdominal ultrasound with liver-focused evaluation
- Alcohol Assessment: Biomarkers for alcohol-related liver disease

Hormonal Health Profiles

Male Hormonal Assessment

- Reproductive Hormones: Total and free testosterone, LH, FSH levels
- Metabolic Hormones: Prolactin, cortisol, growth hormone
- Thyroid Function: Complete thyroid profile
- Stress Assessment: Cortisol rhythm evaluation
- Age-Related Changes: Andropause screening and management
- Performance Optimization: Athletic and wellness hormone evaluation

Female Hormonal Evaluation

- Reproductive Assessment: Estrogen, progesterone, LH, FSH levels
 - Menstrual Cycle Correlation: Cycle-specific timing for optimal interpretation
 - PCOS Screening: Testosterone, DHEA-S, insulin resistance markers
 - Menopause Management: Hormone replacement therapy monitoring
 - Fertility Assessment: Anti-Müllerian hormone, comprehensive fertility panel
 - Stress and Metabolism: Cortisol, thyroid function, metabolic markers
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7. CORPORATE & COMMUNITY SERVICES

Corporate Health Programs

Employee Health Screening Programs

- Customizable Packages: Tailored to company size and industry requirements
- On-Site Services: Mobile laboratory and screening units
- Group Discount Programs: Volume-based pricing for cost-effective screening
- Executive Health Programs: Comprehensive screening for leadership teams

- Occupational Health: Industry-specific screening protocols
- Health Metrics Dashboard: Aggregate health data reporting for HR departments
- Wellness Integration: Coordination with corporate wellness initiatives

Pre-Employment Health Screening

- Comprehensive Medical Evaluation: Complete physical examination and laboratory testing
- Job-Specific Requirements: Customized testing based on job demands
- Fitness Certification: Medical fitness certificates for employment
- Drug and Alcohol Screening: Comprehensive substance abuse testing
- Vaccination Status: Immunization verification and administration
- Rapid Turnaround: Results available within 24-48 hours for urgent hiring needs
- Legal Compliance: OSHA and industry-specific requirement adherence

Occupational Health Monitoring

- Baseline Health Assessment: Pre-exposure health status documentation
- Periodic Surveillance: Regular monitoring for occupational exposures
- Specialized Testing: Lead levels, pulmonary function, hearing assessment
- Regulatory Compliance: OSHA, EPA, and industry-specific requirements
- Medical Surveillance Programs: Ongoing health monitoring for high-risk occupations
- Return-to-Work Evaluations: Medical clearance after illness or injury

Community Outreach Programs

Public Health Screening Initiatives

- Free Community Health Camps: Basic health screening in underserved areas
- Disease-Specific Campaigns: Diabetes, hypertension, and cancer screening drives
- School Health Programs: Student health screening and vaccination programs
- Senior Community Outreach: Geriatric health assessment in community centers
- Health Education: Disease prevention and health promotion workshops
- Partnership Programs: Collaboration with NGOs and government health initiatives

Health Awareness Campaigns

- Preventive Health Education: Community workshops on disease prevention
- Seasonal Health Programs: Vaccination drives, flu prevention campaigns
- Chronic Disease Management: Support groups and educational programs
- Maternal and Child Health: Prenatal and pediatric health promotion

- Mental Health Awareness: Community mental health screening and education
 - Environmental Health: Air quality and water safety testing initiatives
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8. PATIENT SUPPORT SERVICES

Digital Health Services

Telemedicine & Consultation Services

- 24/7 Virtual Consultations: Round-the-clock access to medical professionals
- Specialist Consultations: Direct access to pathologists, radiologists, and specialists
- Report Interpretation: Detailed explanation of test results and clinical significance
- Second Opinion Services: Expert review of complex cases and treatment recommendations
- Follow-up Care Coordination: Ongoing health monitoring and care management
- Emergency Consultation: Immediate medical advice for urgent concerns

Digital Health Platform

- Mobile Application: Comprehensive health management through smartphone app
- Online Portal: Web-based access to health records and test results
- Appointment Scheduling: Real-time appointment booking and rescheduling
- Report Access: Immediate digital delivery of test results
- Payment Processing: Secure online payment with multiple options
- Health Record Management: Lifetime health record storage and access
- Family Account Management: Centralized health management for family members

Home Healthcare Services

Home Sample Collection

- Comprehensive Service: Phlebotomy and sample collection at patient's location
- Qualified Personnel: Trained phlebotomists with proper equipment and safety protocols
- Sample Integrity: Proper handling and transport to maintain sample quality
- Scheduling Flexibility: Same-day and advance appointment scheduling
- Special Populations: Elderly, disabled, and pediatric specialized collection
- Multiple Family Members: Convenient collection for entire families
- Emergency Collection: Urgent sample collection for critical patients

Home Healthcare Coordination

- Care Management: Coordination with home healthcare providers
- Medication Management: Laboratory monitoring for therapeutic drug levels
- Chronic Disease Monitoring: Regular testing for diabetes, kidney disease, heart conditions
- Post-Surgical Care: Laboratory follow-up for surgical patients
- Elderly Care Support: Comprehensive health monitoring for aging population
- Hospice and Palliative Care: Compassionate testing for end-of-life care

Patient Education & Support

Health Literacy Programs

- Test Preparation Guidelines: Detailed instructions for optimal test preparation
- Result Interpretation: Educational materials explaining test results and significance
- Disease Management Education: Resources for managing chronic conditions
- Preventive Health Information: Guidelines for disease prevention and health maintenance
- Medication Interaction Guidance: Information about medications affecting test results
- Lifestyle Modification Support: Diet, exercise, and wellness recommendations

Counseling Services

- Genetic Counseling: Pre and post-test counseling for genetic testing
- Nutritional Counseling: Dietary advice based on test results and health status
- Lifestyle Counseling: Behavior modification for disease prevention
- Psychological Support: Counseling for patients dealing with serious diagnoses
- Family Planning Counseling: Reproductive health and genetic risk counseling
- Chronic Disease Support: Ongoing support for patients with long-term conditions

9. QUALITY ASSURANCE & TECHNOLOGY

Laboratory Accreditation & Standards

International Accreditations

- CAP (College of American Pathologists): Gold standard laboratory accreditation
- ISO 15189:2012: International standard for medical laboratories
- NABL (National Accreditation Board for Testing and Calibration): Indian national accreditation
- JCI (Joint Commission International): Healthcare quality and safety standards
- WHO-AFRO: World Health Organization regional accreditation
- CDC Guidelines: Centers for Disease Control prevention protocols

Quality Management System

- Continuous Quality Improvement: Ongoing process improvement initiatives
- Risk Management: Systematic identification and mitigation of laboratory risks
- Document Control: Comprehensive documentation and version control systems
- Training and Competency: Regular staff training and competency assessment
- Incident Management: Systematic handling of errors and corrective actions
- Customer Satisfaction: Regular feedback collection and service improvement

Advanced Technology Integration

Laboratory Information System (LIS)

- Automated Workflow: Complete laboratory automation from sample receipt to reporting
- Barcode Tracking: End-to-end sample tracking and identification
- Quality Control Integration: Real-time quality control monitoring and alerts
- Result Verification: Multi-level result verification and approval processes
- Critical Value Management: Automatic notification system for critical results
- Data Security: Comprehensive cybersecurity measures for patient data protection

Artificial Intelligence & Machine Learning

- AI-Assisted Diagnosis: Computer-aided interpretation for radiology and pathology
- Pattern Recognition: Advanced algorithms for abnormal result detection
- Predictive Analytics: Risk assessment and outcome prediction models
- Quality Enhancement: AI-driven quality control and error detection
- Workflow Optimization: Machine learning for process improvement
- Decision Support: Clinical decision support tools for healthcare providers

Equipment & Technology

State-of-the-Art Equipment

- Automated Analyzers: High-throughput, precision analytical instruments
- Mass Spectrometry: Advanced technology for precise molecular analysis
- Flow Cytometry: Sophisticated cell analysis and sorting capabilities
- Molecular Biology Platforms: PCR, sequencing, and genetic analysis systems
- Digital Imaging: High-resolution digital pathology and radiology systems
- Point-of-Care Testing: Rapid testing capabilities for emergency situations

Maintenance & Calibration

- Preventive Maintenance: Regular maintenance schedules for all equipment
 - Calibration Programs: Daily calibration and verification procedures
 - Performance Monitoring: Continuous monitoring of equipment performance
 - Upgrade Programs: Regular technology updates and equipment upgrades
 - Backup Systems: Redundant systems to ensure continuous operation
 - Emergency Protocols: Contingency plans for equipment failures
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OPERATIONAL INFORMATION

Service Hours & Accessibility

Laboratory Operating Hours

- Main Laboratory: Monday through Sunday, 6:00 AM to 10:00 PM
- Emergency Services: 24/7 availability for critical testing
- Sample Collection: 6:00 AM to 6:00 PM daily
- Home Collection Services: 6:00 AM to 8:00 PM with advance scheduling
- Radiology Services: 8:00 AM to 8:00 PM daily with emergency availability
- Customer Service: 24/7 phone and online support

Report Delivery Options

- Digital Reports: Immediate online access through patient portal and mobile app
- Email Delivery: Secure encrypted email delivery within hours of completion
- SMS Notifications: Text message alerts for completed reports
- Physical Reports: Available for pickup 24/7 through automated kiosks
- Home Delivery: Optional courier delivery service for physical reports
- Hospital Integration: Direct integration with electronic health records

Patient Support & Communication

Customer Service

- Multi-Channel Support: Phone, email, WhatsApp, and live chat options
- Language Support: Services available in multiple languages
- Patient Advocacy: Dedicated patient advocates for complex cases
- Complaint Resolution: Formal complaint handling and resolution process

- Feedback Systems: Regular patient satisfaction surveys and improvement initiatives
- Educational Resources: Comprehensive health information and testing guides

Emergency & Critical Result Management

- Critical Value Protocols: Immediate notification system for life-threatening results
- Emergency Contact: 24/7 emergency hotline for urgent medical situations
- Hospital Coordination: Direct communication with emergency departments
- Physician Notification: Automatic notification system for referring doctors
- Follow-up Protocols: Systematic follow-up for critical findings
- Documentation: Comprehensive documentation of all critical result communications

This comprehensive service catalog provides detailed information about all services offered by Advanced Medical Testing Lab, designed to serve as a complete reference for patients, healthcare providers, and administrative staff.